



v158TH FIGHTER WING

— HOME OF THE GREEN MOUNTAIN BOYS —



v158 FW / v134 FS / v315 FS

STRIKE!



Air to Air Refueling



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vCOL Emerson "SABRE" C.
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"Setting the Standard Daily!"



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Air to Air Refueling (AAR)



AAR



- **Air to Air Refueling**, or AAR, is a supplementary addition used as a force multiplier, flown for the purpose of transferring fuel from a tanker aircraft to a receiver aircraft during flight. This critical capability allows combat and support assets to remain airborne for extended durations, increasing their operational range, loitering time, and payload capacity by allowing them to take off with less fuel and more ordnance.
- **AAR OBJECTIVES.** The objective of AAR operations is to enhance effectiveness by extending the range, payload or endurance of receiver aircraft. Successful AAR depends on 3 major factors: **Equipment Compatibility, Performance Compatibility, Procedural Compatibility.**
- **Air-to-Air Refueling (AAR).** Event requires tanker rendezvous, hook-up and transfer of fuel or stabilized dry hook-up. More than one event may be credited if receivers accomplish another rendezvous, hook-up and fuel transfer/dry hook-up.



Air to Air Refueling General Procedures & Airspaces



General Proc.



- **Fundamental Principles.** Normally, AAR operations require extensive preplanning to ensure optimum effectiveness whilst maintaining safety and efficiency. This requires the identification of the best tanker type or types for the desired mission set, the selection of the optimum route for the operation, and suitable diversions for the aircraft types. It is essential that an efficient communications interface exist between tanker and receiver flight lead, to ensure the correct positioning and timing of the tanker to meet operational/training demands.
- **SCOPE.** This lesson will address the modes of employment of AAR, the commonality of equipment and identify areas where VATSIM standardization is practicable. Annexes covering specific VSOA procedures have been incorporated where necessary.



AAR Airspaces



- **AAR Areas (AARAs)/Anchor Areas and AAR Tracks**
 - **Airspace Reservations (AARA).** AARAs/anchors areas and AAR tracks are areas of airspace established by the national authority for the conduct of routine AAR training.
 - **Altitude Reservations (ALTRVs):** ALTRVs and military corridors are normally arranged with the appropriate national ATC authorities. For AAR purposes, moving ALTRVs are normally used to guarantee the required route and altitude(s) for an AAR supported deployment.
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Air to Air Refueling Rendezvous Procedures



Rendezvous Proc.



- **Introduction.** The purpose of a RV procedure is to achieve close visual contact between the tanker and a receiver section or element. For the purpose of this Lesson, each RV procedure is written for missions with one tanker. This Section outlines the standard types of RV. The type of RV utilised will be dictated by the mission requirements, available equipment, weather conditions and the EMCON option in force.
- **Altimeter Settings.** Unless otherwise directed, an altimeter setting of 1013.2 mb (29.92 inches) is to be used for AAR operations at or above transition altitude, or when over water and operating in accordance with ICAO procedures. When not operating on standard pressure settings, tanker crews will include the altimeter setting in the RV Initial Call.
- **Vertical Separation.** Receivers are normally to join from below and are to maintain a minimum of 1000 ft vertical separation, unless otherwise stated at the planning or briefing stage, until visual contact and positive identification have been made. Receivers will maintain altitude separation of at least 1000 ft until 1 nm from the tanker.



Rendezvous Proc.



- **Speeds.** Receiving aircraft should normally fly the speed prescribed in AFMAN 11-2F-35AV1 and listed in appropriate tanker National SRD. For Option 1 of the RV Delta (Chapter 2, Annex 2D) where the tanker's speed is known to the receiver, the receiver flies the tanker speed plus 20 kts.
- **Visual With Tanker.** Once the receiver(s) is visual with the tanker, receivers are clear to join and should initiate a progressive climb towards the tanker.
- **Airborne Intercept (AI) Radar.** When possible & national limitations permit, as long as radar lock is maintained, aircraft equipped with an AI radar may continue closure at no more than 10 kts of overtake inside of $\frac{1}{2}$ nm maintaining 500ft vertical separation to a minimum range of 1500 ft



Rendezvous Proc.



- **Visual Contact Not Established.** If visual contact is not achieved at the appropriate minimum closure range, the receiver(s) may:
 - **Stabilize** at the appropriate minimum range and maintain it until the tanker manoeuvres into an area of improved visibility.
 - **Descend** to 1000 ft below the tanker, drop back to 1 nm and either maintain this position until the tanker manoeuvres into an area of improved visibility or terminate the RV.
- **Visual With Tanker.** Once the receiver(s) is visual with the tanker, receivers are clear to join and should initiate a progressive climb towards the tanker.



Rendezvous Proc.



- **Anchor Point.** The refuelling anchor pattern is a left-hand racetrack with legs separated by as little as 7 nm for smaller, the standard leg length for this pattern is 50 nm, as shown in Figure 2A-1. The location of the pattern is determined by the anchor point and the orientation of the inbound course
- **RV Alpha.** The RV Alpha (Anchor) procedure is a RV carried out under the control of a radar station on the ground, in the air or on-board ship. The RV Alpha is normally used to vector receivers to tankers operating on an AARA/anchor area but may be used as required in any situation. However, with the agreement of the tanker, the controller may give the tanker alterations of heading to effect a quicker join-up.
- **PROCEDURE.** The essential requirement for a RV Alpha is positive control by the radar controller to bring the receiver to an ideal position of 1 nm behind and 1000 ft below the tanker.



Rendezvous Proc.



- Receiver Responsibilities:

- FL/Altitude/Height. When directed by the controlling agency, be established at 1,000 ft below the assigned base air refuelling altitude.
- Heading. Fly headings as directed by the controlling agency.
- Receiver Takes Control of RV. Complete the RV using organic AI radar once radar contact with the tanker is established and call: "[C/S] Judy."
- Receiver Visual with Tanker. When visual with the tanker and EMCON procedures permit the use of radios, the receiver calls: "[C/S] Visual," and is then cleared by the tanker to joinup. Initial joinup should occur on the portside of the tanker unless specified otherwise..



Rendezvous Proc.



- **Communication Procedures.** Whilst on station, the tanker will monitor the published AAR frequency. When the controlling agency initiates the receiver RV, it will ensure that the receiver(s) confirm their FL/altitude/height, A/A TACAN (channel), Mode 3 and armament state to the tanker.
 - In EMCON 1, the receiver(s) should not close inside 1 nm until radio contact is established with the tanker.
 - If either the tanker or receiver(s) is not at its briefed FL/altitude/height an additional radio call is to be made when established at its nominated FL/altitude/height.
 - During EMCON 2, radio calls will not be made unless they are necessary to ensure safe vertical separation.
- **Vertical Separation.** The RV vertical separation is to be maintained until 1nm from the tanker(s) and visual contact is established. The receiver(s) will then commence a gradual climb to the astern if the tanker(s) is not acquired visually by 1 nm.



Rendezvous Proc.



- **RV Golf (En Route)** The RV Golf facilitates a join up en-route on a common track to make good a scheduled time to join an ALTRV or other established military corridor; the tanker(s) and receiver(s) may have departed either from the same or different bases
- **Communication Procedures.** Fifteen minutes prior to the ARCT the tanker and receiver(s) are to confirm their FL/altitude/height, Mode 3, armament state and timing. The receiver(s) should fly towards the ARCP with A/A TACAN and radar beacon on (if appropriate) at 1000 ft below the base AAR altitude.
- **Visual Acquisition of Tanker(s)/Receiver(s).** When established on the common track, tanker(s) and receiver(s) are to use all available locating aids (EMCON state permitting) to gain visual contact between the tanker(s) and the receiver(s).



Air to Air Refueling Joining



Joining Proc.



- **Join - Single Tanker – Echelon Left Position.** Receivers should plan to arrive from tanker's left side, unless the lead tanker directs otherwise, and form up in the Echelon Left position. The Echelon Left position is behind the wingline of the tanker. See Figure 2-14 and Figure 2-25A.
- **First Boom Receiver.** Once the boom operator has a good rear view of the receiver and is able to give detailed commentary and advice to both the receiver and the tanker. If cleared by the tanker, the first receiver of a formation may join directly behind the boom. The receiver visually must confirm that no AAR is in progress and that the boom is lowered. See Figures 2-5B.
- **Remainder of Formation.** All other members of the formation should form up in the Echelon Left position. See Figures 2-5A and B.



Joining Proc.



- **Receiver Sequence.** To ensure safe operations, the tanker will direct receivers to the formation position that they are to adopt, i.e. the Echelon Left, refuelling or Echelon Right. Furthermore, the tanker will direct when receiver aircraft are to reposition, and then only one aircraft at a time is to be changing formation position; all other aircraft should remain steady in the formation position allocated by the tanker; this command may be given by radio or by using the standard procedural signals in EMCON conditions. See Figure 2-5 for illustrations of controlled receiver flow around the tanker.
- **Collision Avoidance.** Receivers are responsible for ensuring that the airspace they are moving into is clear of other aircraft. In addition, moving receivers are mutually responsible for ensuring that they do not collide with other repositioning aircraft.
- **Remainder of Formation.** All other members of the formation should form up in the Echelon Left position. See Figures 2-5A and B.



Joining Proc.



- **Refuelling First Receiver.** The first receiver is refuelled and, when complete moves to the Echelon Right position on the right wing.
 - **Refuelling Complete.** When fuelling is complete, the valves are closed and the boom is automatically or manually retracted. Receiver aircraft should call "Disconnect". Once cleared from behind the boom, receiver will reform in Echelon Right Position.
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Air to Air Refueling Leaving



Leaving Proc.



- **Leaving.** Once refuelling is complete, receivers will be cleared to the Echelon Right position. If there are two or more receivers, they should reform using Echelon Right formation, moving sequentially outboard of the tanker with the first receiver remaining closest to the tanker's wing. From this position, they leave the tanker either level with the tanker or climbing.
 - **Vertical Separation.** Normally, receivers must not descend from a tanker formation unless co-ordinated with an ATC or other controlling unit as other receivers are likely to be joining the tanker from below.
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Air to Air Refueling Comms/IFF



Comms/IFF



- **IFF/SIF.** IFF/SIF is to be switched to STBY at the “Visual” call and is to remain off for the duration of AAR, until the flight leaves the formation.
- **CTR Join Comms Flow:**
 - “[C/S] Request Vectors to [Tanker C/S] for Rejoin, RV Alpha”.
 - “[C/S] Request Entrance into AR-609 to rejoin with [TKR C/S] RV Alpha.
 - “[C/S] Would like to activate our ALTRV (ALTRV Alt) and wait for [TKR C/S] at [WYPT]
 - “[C/S] Would like to Enter the ALTRV to rejoin with our tanker [Tanker C/S]”

Ensure to Declare MARSA w/ [Tanker C/S] no later than 5nm

- **Tanker RV Comms flow.**
 - “[Tanker C/S] [C/S] 15nm Request Rejoin.”
 - “[Tanker C/S] [C/S] 5nm W/ Sensors”
 - “[Tanker C/S] [C/S] 2nm Visual, Nose cold switches safe”



Comms/IFF



- **Leaving Comms flow.**
 - “[Tanker C/S][C/S] Clearing High Right/Left
 - “[Tanker C/S][C/S] Switching Common”

- **Center Tanker RV Comms flow.**
 - “[C/S] Leaving [Tanker C/S] We’d like to discontinue MARSA”



Additional Study Material

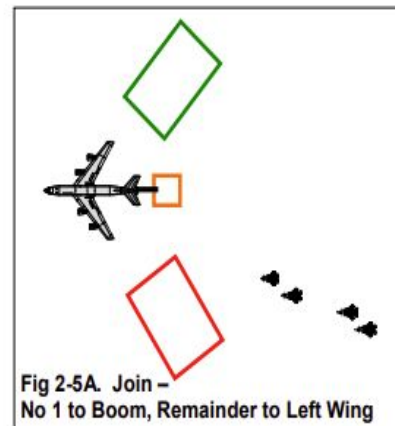
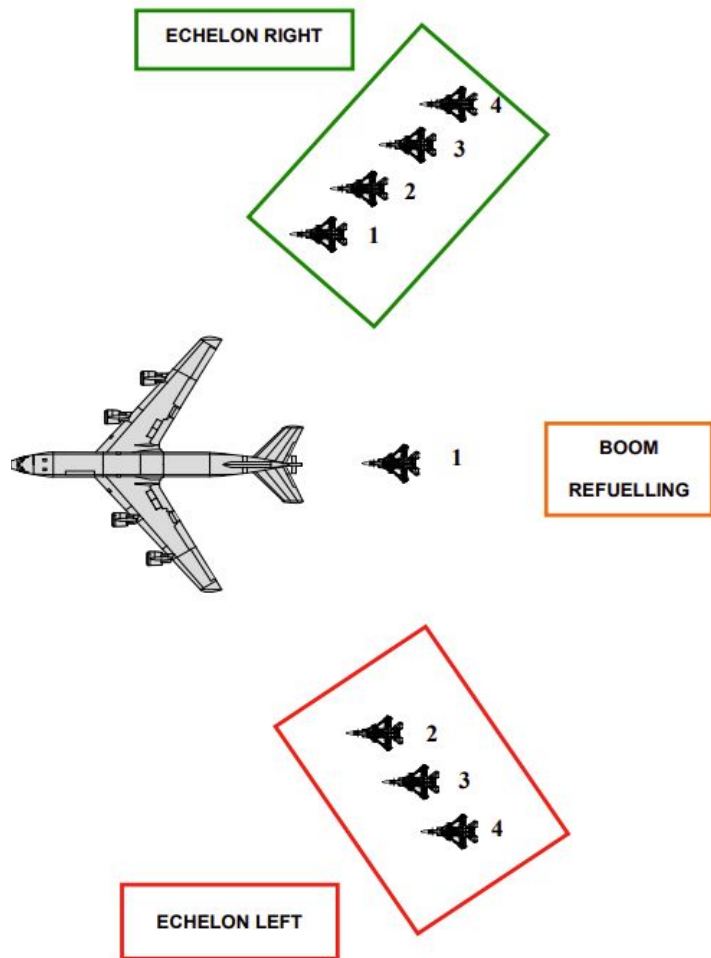


Fig 2-5A. Join - No 1 to Boom, Remainder to Left Wing

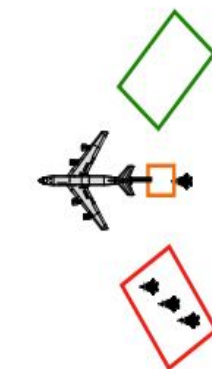


Fig 2-5B. Echelon Left



Fig 2-5C. Receiver Refuels

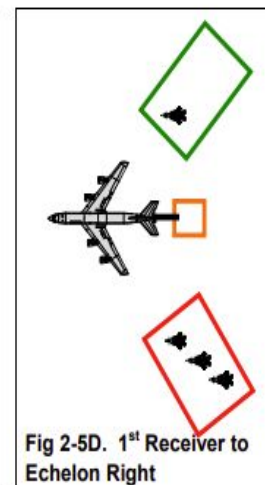


Fig 2-5D. 1st Receiver to Echelon Right



Fig 2-5E. 2nd Receiver Refuels



Additional Study Material

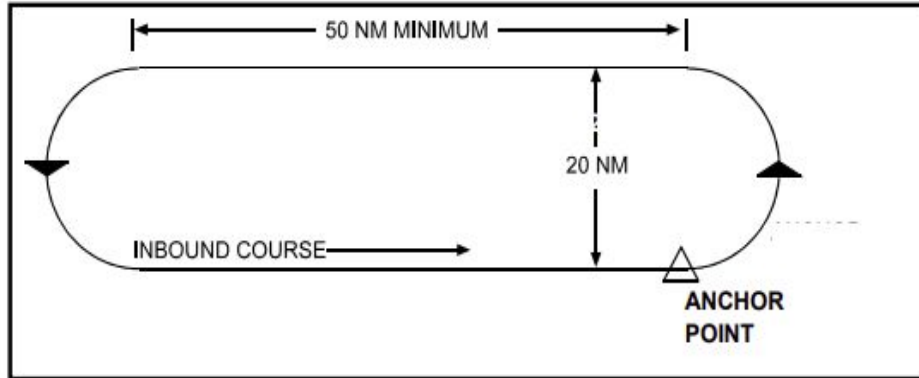


Figure 2A-1. Anchor Pattern

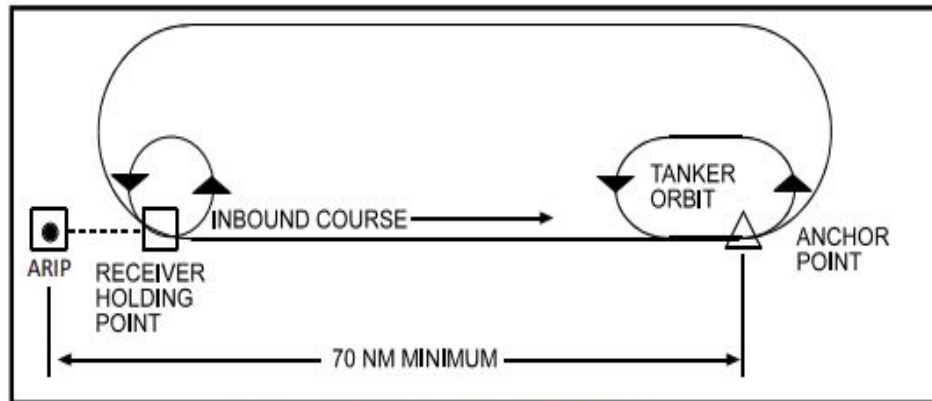


Figure 2A-2. Alternate Anchor Pattern

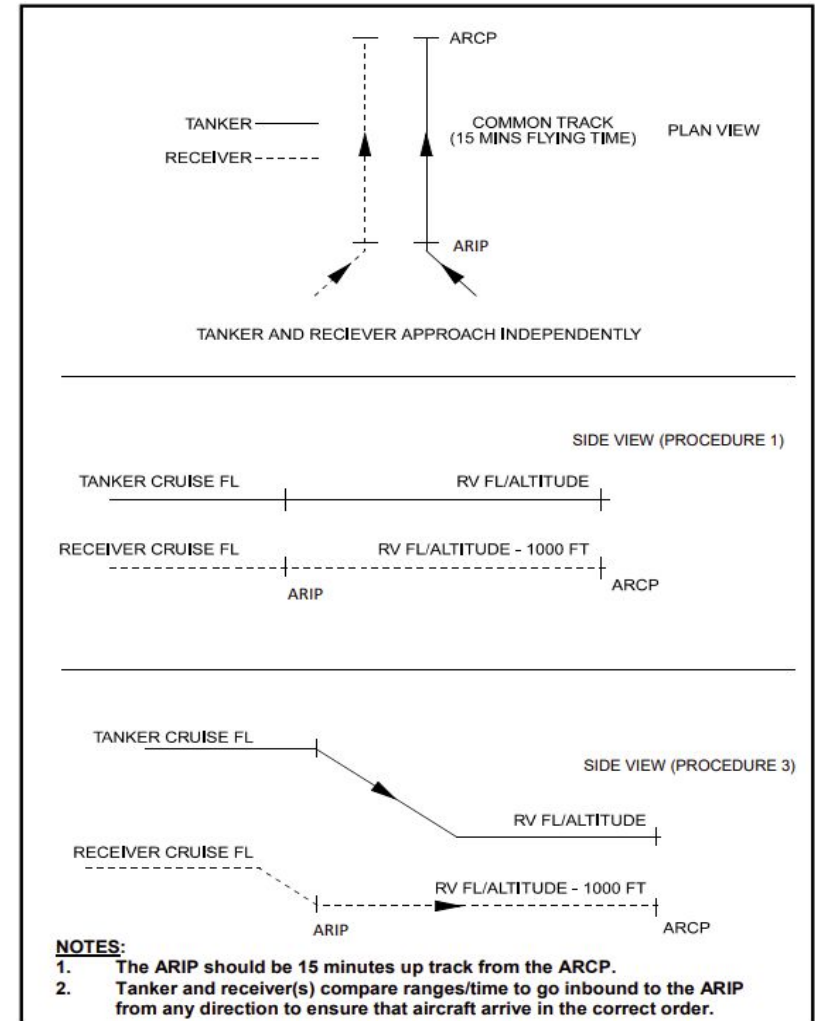


Figure 2G-1. Diagram of RV Golf (En Route)



End



You have completed this lesson portion covering AAR.

